MACHINE LEARNING & ALGORITHM R&D

Xander Xu

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Algorithm engineer with experience in image segmentation, probabilistic modeling and biomedical prediction; co-first author of two peer-reviewed papers. Recent work covers tool-augmented LLM inference, self-play policy experiments and reliable research execution.

Technical capabilities

Modeling	Python, PyTorch; U-Net, convolutional features, GMM, feature selection
LLM methods	vLLM; tool-augmented reasoning, self-consistency, answer aggregation; RAG, LoRA adaptation
Policy experiments	BC/PPO experimentation, JAX training configuration, paired evaluation, NumPy deployment

Selected technical projects

AIMO3 · Tool-augmented mathematical reasoning

2026 · Kaggle bronze · 405 / 4,138 teams

- Modified a public tool-integrated reasoning baseline with weighted answer voting, using Python-call counts and error rates as confidence heuristics.
- Ran inference experiments with gpt-oss-120B/vLLM, eight parallel attempts, persistent Python execution, consensus early stopping and dynamic time allocation.

Orbit Wars · Self-play policy experiments

2026 · Kaggle silver · 112 / 4,729 teams

- Designed a comeback-gated dense-reward experiment on an existing BC→PPO self-play stack; adjusted JAX training configuration and compilation caching.
- Combined 2P/4P policy routing, NumPy deployment and paired evaluation, rejecting reward candidates that failed evaluation gates.

Recent experience

Jul 15, 2025 – Present
China

ChanceTop · AI Engineer

- Contributed to Core-AI workflow persistence and pause/resume; fixed stale state overwrites during cancellation and cross-pod execution-state misclassification, with regression coverage.
- Implemented capability-aware model routing for reasoning and tool use across OpenAI-compatible and Responses APIs.
- Applied agent workflows to structured data processing and Local SEO automation.

Sep 2024 – Jul 2025
Xiamen, China

Qudian Technology · LLM Application Engineer

- Designed multimodal agent workflows combining retrieval, function calling, image generation and speech interfaces, with prompt and content-safety controls.

Mar 2023 – Jun 2024
Xiamen, China

Jiyuan Technology · Technical Lead, Medical LLM R&D

- Integrated an oncology knowledge base through RAG, applied LoRA for domain adaptation, and developed intent-recognition and entity-extraction workflows.

Earlier algorithm & research experience

Apr 2022 — Mar 2023
Shenzhen, China

Fapon Biotech · Algorithm Engineer and Project Lead

- Led algorithm design for recurrence-risk stratification from whole-slide pathology images.

Apr 2021 — Apr 2022
Shenzhen, China

BGI Research · Algorithm Engineer and Team Lead

- Developed attention-enhanced semantic segmentation for ssDNA-stained tissue images.
- Proposed GMM-based Stereo-seq correction combining spatial and expression information; contributed to the associated PCT patent application.

Oct 2018 — Apr 2021
Xiamen, China

Jiyuan Technology · Algorithm Engineer

- Developed H&E cell segmentation and classification using U-Net and convolutional features including ResNet, DenseNet and Siamese networks.
- Built metagenomic geolocation with feature selection, classification and interpolation; contributed expression-feature and machine-learning analysis to anti-PD-1 response research. Co-first author on both studies.

Jul 2017 — Oct 2018
Beijing, China

Tsinghua University, Institute for Interdisciplinary Information Sciences · Research Assistant

- Built heterogeneous-network features for drug–target ranking and applied deep-learning detection to low-signal cryo-EM images.

Selected research

1. **Ratio of the interferon-γ signature to the immunosuppression signature predicts anti-PD-1 therapy response in melanoma**
   npj Genomic Medicine 6, 7 (2021) · Co-first author
2. **A machine learning framework to determine geolocations from metagenomic profiling**
   Biology Direct 15, 27 (2020) · Co-first author
3. **Inconsistent prediction capability of ImmuneCells.Sig across different RNA-seq datasets**
   Nature Communications 12, 4167 (2021) · Co-author

Competitions & technical communication

Kaggle Notebooks Master · 6 gold · 4 silver notebook medals
Competitions Expert, with a further 2 silver and 5 bronze competition medals across modeling and agent challenges.

过拟合青年 · Bilibili · 24,091 followers · Animated explanations of AI mechanisms
Updated 29 Oct 2026

Education

Fujian Medical University · BSc in Bioinformatics · 2013 — 2017